

A Fraction of a Whole Number or Set

Finding one equal group as a unit fraction of a set, finding several equal groups as a non unit fraction, and working backwards from a part to the whole

The two moves, every time.

- **Divide by the bottom number.** It says how many *equal groups* the whole set is broken into. One group is the unit fraction.
- **Multiply by the top number.** It says how many of those groups to take.
- So $\frac{2}{5}$ of 20 means: make 5 equal groups of 4, then take 2 of them. Divide first, multiply second.
- Draw or circle the groups before you write any numbers — the picture is what makes the two moves obvious.

1. A box holds 24 crayons. Ravi takes $\frac{1}{4}$ of them. Circle the crayons in the picture to show the equal groups, then write how many crayons Ravi takes.



Answer: _____

- 2.** Nia has 21 stickers laid out below. She gives $\frac{1}{3}$ of them to her brother. Ring the equal groups, then write how many stickers her brother gets.



Answer: _____

- 3.** A jar holds 18 marbles. Sofia takes $\frac{2}{3}$ of them. Use the bar below: split it into the number of equal parts the bottom number asks for, write how many marbles are in one part, then shade the parts the top number asks for. How many marbles does Sofia take?



Answer: _____

4. A book has 32 pages. Dev has read $\frac{3}{4}$ of it. Split the bar into equal parts, label one part, and shade what Dev has read. How many pages has he read?

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Answer: _____

5. Now work backwards. Leo ate $\frac{1}{5}$ of a bag of grapes, and that one part was 8 grapes. The bar below is the whole bag, already split into the right number of equal parts. Write 8 in each part. How many grapes were in the whole bag?

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Answer: _____

6. An activity period lasts 42 minutes. The class spends $\frac{5}{6}$ of the period on a project and the rest tidying up. How long do they spend on the project? Give your answer in minutes.

Answer: _____ minutes

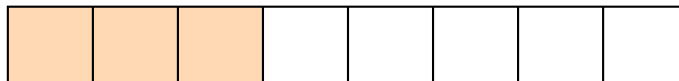
7. A jug holds 2.4 litres of lemonade. Priya pours $\frac{1}{8}$ of the jug into a glass. How many litres go into the glass?

Answer: _____ L

8. There are 28 students in a class, and $\frac{3}{7}$ of them ride the bus to school. How many students do *not* ride the bus? Read the question twice — it does not ask for the same number the fraction gives you.

Answer: _____

9. A theatre is $\frac{3}{8}$ full, and that is 96 people. The bar below is the whole theatre, split into eighths, with the full part shaded. Work out how many seats one eighth holds, then find how many seats the theatre has altogether.



Answer: _____

10. Sam is asked for $\frac{3}{5}$ of 45 counters. He splits the 45 counters into five equal groups, points at one group, and says the answer is 9. Sam has done part of the work correctly. Write one sentence saying what he did right and what he forgot, then give the correct answer.



Answer: _____